

8 **(short)** A mechanism consisting of two uniform bars is held in place by a linear spring of spring constant k . Points A and C are fixed, while B is constrained to move horizontally as shown in Fig. 8. Each bar has mass m and length L . The spring is unstretched when the angle α is 0° , and the system is in equilibrium when α is 45° .

- (a) Obtain an expression for the potential energy of the system as a function of the spring extension x . [5]
- (b) Determine a value for the spring constant k . [5]

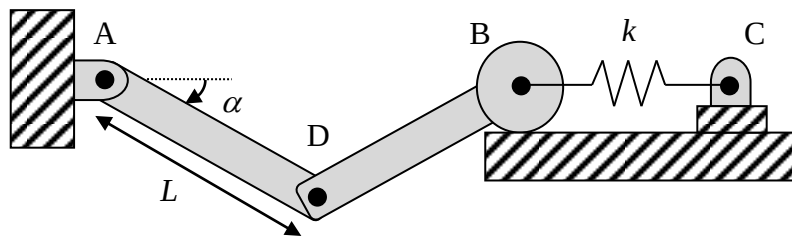


Fig. 8